KUSHAL RUPSING MUDE

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EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering — **Currently in Third Year**
Sinhgad College of Engineering, Vadgaon (BK) — Savitribai Phule Pune University (SPPU)
CGPA: 9.55 (FE) | **9.18** (Sem 3)

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript (ES6+), SQL, HTML5, CSS3
- **Web & Full-Stack:** React.js, Next.js (App Router), Node.js, Express.js, MongoDB, Tailwind CSS
- **AI & Machine Learning:** Google Gemini API, PyTorch, Scikit-learn
- **Developer Tools & Core:** Git, GitHub, Supabase, OOP, DBMS, OS, Computer Networks

PROJECTS

Metrica – Digital Instrument Trust & Verification Platform | [Live Demo] [GitHub] — **GovTech & Enterprise**
Next.js 15, React 19, TypeScript, Prisma ORM, Supabase / PostgreSQL, Leaflet GIS, Tailwind CSS, Zod
- Engineered an automated lifecycle audit & verification system for the Department of Consumer Affairs (DoCA, PS 26036).
- Designed tamper-proof digital certificates, cryptographic QR auditing, and interactive GIS mapping for on-ground field inspections.
- Implemented secure role-based access control (RBAC) and robust PostgreSQL database schemas with Prisma ORM.

Antarctic AI (POLARIS-X) – Multimodal Intelligence Assistant | [Live Demo] [GitHub] — **AI & Gemini API**
Google Gemini API, Python, React.js, Node.js, Express.js, MongoDB, Tailwind CSS
- Built an intelligent multimodal reasoning assistant integrating Google Gemini API with custom data parsing pipelines.
- Engineered streaming query architecture achieving low-latency first-token responses (<80ms) across multi-turn sessions.
- Developed high-performance backend microservices for contextual document query parsing, memory storage, and analytics.

Code Among Us (CodeMongUs) – Multiplayer Real-Time Arena | [Live Demo] [GitHub] — **Multiplayer & Real-Time**
React.js, Node.js, Express.js, Socket.io, Monaco Code Editor, Tailwind CSS
- Created a real-time multiplayer coding platform inspired by Among Us, fusing social deduction mechanics with competitive coding.
- Integrated Monaco Editor for in-browser syntax evaluation, live execution, and bi-directional real-time room synchronization via Socket.io.
- Optimized event broadcasting to handle concurrent player actions with sub-50ms latency.

HACKATHON EXPERIENCE & LEADERSHIP

Smart India Hackathon / DoCA Track (2026) | *Team: Kushal-Mangal*
- Selected for Problem Statement 26036 (Department of Consumer Affairs); led full-stack system architecture and deployment for Metrica.

Velora Hackathon | *Team: Fourtified Minds*
- Developed 'Code Among Us'; architected game logic and real-time state synchronization for live coding puzzles under 24-hr constraints.

AI for Bharat Hackathon | *Team: Bit Beast*
- Built 'Comic Forge' — text-to-comic multimodal prototype generating visual storytelling scenes with AI models.
- Active participant in collegiate and national hackathons, leading system architecture, rapid prototyping, and presentation.

CERTIFICATIONS

- **Certificate of Participation** — Smart India Hackathon / DoCA Track
- **Certificate of Participation** — Velora Hackathon (Team Fourtified Minds)
- **Certificate of Participation** — AI for Bharat Hackathon (Team Bit Beast)